

ANDREW MAZZOLA

andrewmazzola.com · github.com/rust4ai

WORK EXPERIENCE

Principal Engineer

June 2021 – Present

Series A Fintech (NDA) · Remote — Backed by Toyota Ventures & Franklin Templeton

- Lead engineer and DevOps for financial SaaS, architecting and deploying production lending products using Rust and TypeScript across AWS and DigitalOcean
- Architected an agentic test harness for the company's AI agent, validating nondeterministic LLM behavior and fixing critical issues in integer handling, edge cases, and tool call parsing
- Own end-to-end implementation lifecycle from architecture through production deployment, monitoring, and iteration

Forward Deployed Engineer

Oct. 2014 – June 2021

McNaughton-McKay Electric Company · Madison Heights, MI

- Forward deployed across 100+ manufacturing customers, traveling 6,000+ miles/year for on-prem deployments, integration, commissioning, and production handoff
- Led deep discovery on customer engineering environments, defined success criteria, and architected end-to-end automation solutions tailored to each facility's existing stack
- On-site programming and startup for Allen-Bradley Kinetix robotics, 200+ HP VFD, and Studio 5000 PLCs using EtherNet/IP, ControlNet, DeviceNet, and other industrial networks
- Built production-grade internal tooling: call logging system and CRM replacing ad-hoc processes, with user login and lead generation for sales

AGENTIC CONSULTING & OPEN SOURCE

Metalcraft — Agent Orchestration Framework [rust4ai/metalcraft](#)

- LangGraph-style stateful graph orchestrator for AI agents in Rust: typed state, cyclic workflows, human-in-the-loop interrupts, parallel execution, async streaming
- Pluggable persistence, tool registry (Anthropic + OpenAI formats), before-tool hooks, step guards, Mermaid visualization — published crate

Spice — AI Agent Test Harness [rust4ai/spice](#)

- Test framework for nondeterministic LLM agents: 30+ assertions, security allowlists, M-of-N consensus modes, JSON trace recording
- Provider-agnostic interface for OpenAI, Anthropic, or custom implementations

Inference Proxy Server [rust4ai/inference-proxy-server](#)

- Secure LLM proxy for organizations via Amazon Bedrock: scoped API keys, audit logging, OpenAI-compatible endpoint, Docker Compose + PostgreSQL

Metalcraft Agent — Reference Agent App [rust4ai/metalcraft-agent](#)

- Modular agent application built on Metalcraft + Spice: persona-based execution, interactive and task modes, tool auto-approval workflows

StarkBot CLI — Terminal AI Agent [starkbotai/starkbot-cli](#)

- Terminal-based AI agent platform in Rust (Ratatui): 10 integrated tools (bash, file ops, grep, web fetch, sub-agent spawning), four tabbed views (Chat, Skills, Graph, Memory)
- Four agent personas (coding, research, director, devops) with per-tool approval hooks, error spiral detection, and output truncation safety
- SQLite-backed memory with full-text search and embeddings, automatic token estimation and sliding-window context compaction

Vix Automation — [Starflask.com](#)

- Multi-tenant agentic inbound voice SaaS: custom workflows, API integrations, logical filtering and branching

Mod Technologies — 2026

- Custom agentic workflow for media team: auto-compress, transcribe, summarize video via OpenAI. Saves 5+ hrs/week, generates blog posts and thumbnails

ADDITIONAL OPEN SOURCE

- **rust4all-template**: Full-stack web template (Axum + React + PostgreSQL) — JWT auth, rate limiting, security headers, Docker, compile-time SQL · **knowledgebase-agent**: Vectorless RAG via BM25 PageIndex

TECHNICAL SKILLS

Languages	Rust, TypeScript, Python, Shell scripting
AI / Agents	Agent orchestration (stateful graphs, multi-agent), prompt engineering, tool use, RAG, nondeterministic testing, LLM proxy, Anthropic & OpenAI APIs
Infrastructure	Docker, Kubernetes, AWS, DigitalOcean, Railway, CI/CD, GitHub Actions, Linux
Web / Data	Axum, Actix, React, WebSockets, REST APIs, OAuth/OIDC, PostgreSQL, SQLite, SQLx

EDUCATION

B.S. Electrical Engineering — Kettering University, Flint, MI — Magna Cum Laude

Oct. 2014

President, Alpha Gamma chapter of Phi Gamma Delta · EKN Honor Society